SeungEun Chung

PERSONAL DETAILS

Name:	Seungeun Chung
Designation:	AI Developer
Date of Birth (yyyy/mm/dd):	2001/02/22
Email:	SeungeunC@fpt.com

EDUCATIONAL BACKGROUND

- Kookmin University(Bachelor of Electronic Engineering), 03/2020–02/2024
- ● Kookmin University(Master of Electronic Engineering), 03/2024–02/2026

CERTIFICATES

PROFESSIONAL SUMMARY

- Exoerienced in Python, PyTorch, and overall AI model development.
- Experienced in handling multimodal AI and reasoning segmentation via MLLMs.
- Familiar with Vision AI tools.
- Have experience with Prompt Engineering (CoT, In-Context Learning)
- Have experience with real-world data pipelines and edge device optimization

TECHNICAL SKILLS

Skill	Proficiency	Experience
Language		
Python	Advanced	3y
Framework/Library		
PyTorch	Intermediate	3y
OpenCV	Intermediate	3y
Hugging Face	Intermediate	2y
Numpy	Intermediate	3y
Pandas	Intermediate	3y
Tools		

Git	Intermediate	3y
Docker	Fundamental	2y
Operating Systems		
Linux	Fundamental	3y

LANGUAGE SKILLS

- English General Professional Proficiency
- Korean Native

PROFESSIONAL EXPERIENCE

- Total work experience (Year/Month) : 3/2

Project 1	
Duration	05/2025 - 03/2026
Team Size	2
Position	AI Researcher
Description	Research and development of a training-free Vision-Language framework for Reasoning Segmentation. The project addresses the "intent gap" in Referring Image Segmentation (RIS) by utilizing MLLMs as an external reasoning brain, converting complex implicit instructions into explicit visual queries without fine-tuning the segmentation model.
Country/Industry	Korea/AI Research
Business Domain	Artificial Intelligence
Responsibility	• Defined the core problem and literature review of MLLM-based reasoning segmentation. • Developed a pipeline translating complex text instructions into executable queries • Applied Chain-of-Thought (CoT) prompting to enhanced logical reasoning. • Implemented In-Context Learning to improve performance without fine-tuning. • Analyzed spatially aware semantic segmentation behavior in MLLMs.
Technology	• Python, PyTorch • MLLM • Prompt Engineering • Docker

Project 2	
Duration	09/2023 - 04/2025
Team Size	3
Position	AI Researcher
Description	Development of 'GuiFiR', a novel weakly and semi-supervised learning framework for joint hand and guitar fretboard pose estimation. By integrating audio-transcribed pseudo-GT tablature and synthetic data, the project overcomes severe finger occlusions inherent in vision-only guitar playing scenarios
Country/Industry	Korea/AI Research
Business Domain	Artificial Intelligence
Responsibility	● Constructed the GuiFiR-Synth and GuiFiR-Vid datasets, utilizing Grounding DINO to automate bounding box generation for 5,800 images. ● Designed a weakly supervised learning pipeline applying pseudo-GT tablature from an audio-transcription model (TabCNN) and the Hungarian matching algorithm. ● Integrated and fine-tuned state-of-the-art 3D Hand Pose Estimators (HaMeR, WiLoR) and Fretboard Pose Estimator (MMPose). ● Achieved State-of-the-Art (SOTA) performance with an MAE-H of 9.372 on the proposed multimodal task.
Technology	● Python, PyTorch ● MMPose, HaMeR, WiLoR, TabCNN ● GroundingDINO

Project 3	
Duration	09/2024 - 02/2025
Team Size	4
Position	AI Engineer
Description	Design and implementation of a real-time object detection system optimized for high-resolution (4K) video streams in mobility environments. The core focus was to develop a lightweight vehicle license plate detection pipeline deployable on resource-constrained edge computing devices.
Country/Industry	Korea/Automobile

Business Domain	Mobility
Responsibility	● Analyzed industry partner's needs for high-resolution (4K) real-time processing on resource-constrained edge hardware. ● Designed a frame partitioning algorithm to bridge the gap between model accuracy and hardware limitations, improving F1-score from 88.25% to 93.48%. ● Independently built a DeepStream-based pipeline and delivered the end-to-end system on schedule.
Technology	● Python ● Object Detection (YOLO) ● Jetson Orin NX

Project 4	
Duration	09/2023 - 12/2023
Team Size	5
Position	AI Developer
Description	An industry-academic collaborative project aimed at developing a personalized healthcare and beauty AI solution. The project focused on building a vision-based diagnostic module capable of predicting skin health conditions by correlating clinical facial images with microbiome data.
Country/Industry	Korea/Healthcare
Business Domain	Healthcare
Responsibility	● Conducted regular review meetings with the partnering company to align technical model outputs with the business goal of microbiome-based skin scoring. ● Implemented Grad-CAM visualization to provide clear, explainable diagnostic evidence (XAI), ensuring client trust. ● Achieved an average accuracy of 82.57%, surpassing the 80% target, and successfully delivered the complete module.
Technology	● Python ● Grad-CAM

Project 5	
Duration	03/2023 - 07/2023
Team Size	3

Position	AI Developer
Description	Conceptualization and development of an assistive technology solution designed to enhance the social interaction capabilities of the visually impaired. The project involved creating a wearable vision AI system that detects real-time facial expressions of conversational partners.
Country/Industry	Korea/Healthcare
Business Domain	Assistive Technology
Responsibility	● Designed and evaluated a wearable system based on usability and comfort, proving the ability to interpret technical functions into user-centered benefits. ● Awarded Grand Award (1st Place) at ASCC 2023 (Bangkok) for innovation and clarity in global research communication.
Technology	● Python ● Arduino Uno